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Idea Submission · Report Document

iTantra

Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for Low Bitrate Links

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SECTION 1

PROPOSED SOLUTION

1.1 Detailed explanation of the proposed solution

1.1.1 The physical fact the solution is built around

Communication is the first casualty of a disaster. When mobile towers fail, the equipment that continues to work is the handheld radio set, because it requires no infrastructure at all — only a battery and an antenna. Such a radio provides a narrowband data channel of a few hundred bits per second, fixed by terrain, power budget and spectrum allocation. A voice call cannot cross it.

To reproduce a human voice with enough fidelity that words remain distinguishable, a microphone must sample air pressure sixteen thousand times per second, each sample sixteen bits wide. That is 256 000 bits for every second of speech. Opus, the most capable open-source speech codec available, reduces this to approximately 6 000 bits per second at the cost of a thin, robotic timbre; below that rate intelligibility collapses. The gap between that floor and the capacity of a narrowband link remains a factor of twenty.

This is not a software problem that a cleverer codec will eventually solve. It is a *bound*. Audio codecs compress the waveform — the shape of the pressure wave — and a waveform detailed enough to be understood has an irreducible size. One can approach that bound; one cannot cross it by a factor of twenty.

```
       100          1 000        10 000       100 000     1 000 000   bits/second
        |------------|------------|------------|------------|
  raw PCM speech                                     256 000 |=============|
  Opus practical floor                        6 000 |======|
  narrowband radio link       300 |==|
                                   ^
                                   +-- nothing to the right of this can cross the link
```

Figure 1.1: *Data rates on a logarithmic scale. Both audio representations lie far beyond the capacity of the link that survives a disaster.*

1.1.2 The move that changes the problem

The proposed solution stops transmitting the waveform. The sentence *"two people injured, we need a boat, sector seventeen"* carries perhaps forty bytes of actual information. Everything else in those three seconds of audio — the speaker's timbre, the wind behind him, the exact pitch contour, the room acoustics — is not the message. It is the carrier the message arrived in.

If the handset in the sender's hand converts speech to text before transmission, and the handset at base converts the text back to speech after reception, then what crosses the radio link is forty-four bytes instead of ninety-six thousand. The two humans still only ever speak and listen. Neither of them reads or types anything. Between them, the message travels as language rather than as sound.

iTantra is therefore an Android application that turns two ordinary handsets — or a handset and a radio module — into a walkie-talkie in which speech is recognised on the sending device, transmitted as a few dozen bytes, and re-synthesised on the receiving device. Ten Indian languages are supported: Hindi, Gujarati, Marathi, Kannada, Malayalam, Tamil, Telugu, Odia, Bengali and English. The system runs entirely offline, with no cloud service, no SIM, no proprietary software development kit, and no internet permission declared in the application manifest.

1.1.3 Why the interface must remain voice at both ends

A reasonable objection arises: if the payload is text anyway, why not simply let people type it? The problem statement answers this directly, and the answer is the reason the project exists rather than merely being clever.

- **Literacy.** A system that requires typing excludes the people most likely to need it. In a disaster the person holding critical information may be a fisherman, a farmer or a village elder, not necessarily someone who can compose a message in a script under stress.
- **Hands and eyes.** A soldier at altitude wearing gloves, a rescuer on a rope, a driver — none of them can type. All of them can press one large button and talk.
- **Speed under stress.** Speaking a sentence takes three seconds. Typing it in Devanagari on a mobile keyboard, in darkness, with cold hands, takes forty — if it happens at all.
- **Language.** Ten Indian languages and ten scripts. Voice is the only interface equally native to all of them.

The interface is therefore voice, because voice is inclusive. The transmission is text, because text fits. iTantra is the machine that sits between the two.

1.1.4 Three levels of compression

The system does not use a single compression scheme. It degrades gracefully through three, choosing the most aggressive that the situation allows.

Level 1 — Semantic transcoding. Speech becomes UTF-8 text. This is where the enormous ratio originates: roughly 770 to 1 before any further work. Everything below is refinement.

Level 2 — Script packing. UTF-8 encodes Devanagari, Tamil, Kannada and the other Indic scripts at three bytes per character, because UTF-8 must be capable of representing every script simultaneously. However, the frame header already declares which language the message is in. Given that, the receiver knows exactly which 128-codepoint Unicode block applies, and subtracting the block base yields a single byte per character. Three bytes become one. The transformation is lossless, table-driven, costs microseconds, and is available only because we control both endpoints. Characters lying outside the declared block — Latin digits, punctuation, and the zero-width joiners that govern conjunct formation in Indic scripts — are carried by a four-byte escape sequence, so that the transformation remains lossless on ordinary input.

Level 3 — Template codes. In an emergency, vocabulary is small and predictable. A shared table maps the most common operational sentences to single-byte identifiers.

```
0x01   We need medical assistance
0x02   Fire — evacuate immediately
0x03   Position secure, no casualties
0x04   Send a boat
0x05   Request immediate extraction
...    up to 255 entries per deployment profile
```

Figure 1.2: *Extract from the template table.*

After recognition the resulting text is fuzzy-matched against this table. On a confident match the system transmits the identifier rather than the sentence — a single byte. The receiver's table renders the full sentence and speaks it. On a weak match it falls back to Level 2 automatically, and the user is never aware of the switch.

Representation	Size	Bits	On a 300 bps link	Note
Raw PCM, 16 kHz, 16-bit mono	96 000 B	768 000	43 min	What the microphone produces
Opus at 16 kbps	6 000 B	48 000	2.7 min	Good quality voice call
Opus at 6 kbps	2 250 B	18 000	60 s	Thin, robotic, the practical floor
UTF-8 text	99 B	792	2.6 s	What the recogniser produces
+ framing and addressing	111 B	888	3.0 s	12-byte header and trailer
+ script packing (Level 2)	**44 B**	352	**1.2 s**	One byte per Indic codepoint
Template code (Level 3)	**13 B**	104	**0.3 s**	Header plus a one-byte identifier

Table 1.1: *A representative three-second Hindi sentence (हमें तुरंत मदद चाहिए, दो लोग घायल हैं) in each representation.*

1.1.5 What is discarded, and how it is compensated

Honesty about the trade-off is a strength rather than a weakness, and every loss below has a designed answer.

What is discarded	Why it matters	Compensation designed into the system
Speaker's voice identity	Every message would sound like the same synthetic voice	A one-byte sender identifier in the header, a display name in the interface, and a distinct synthesis voice for each node using multi-speaker embeddings
Emotion, urgency, prosody	A shout and a whisper produce identical text	An explicit priority class; alert traffic is routed to the alarm stream at maximum volume and synthesised with a faster, harder delivery profile
Background audio	The listener loses situational cues such as water or machinery	Accepted. Out of scope, and arguably desirable given the bandwidth budget
Anything the recogniser gets wrong	An error is unrecoverable; the listener has no audio to fall back upon	Confidence scoring transmitted with every frame, on-screen display of the recognised text before transmission, domain vocabulary biasing, and mandatory confirmation for alert-class messages

Table 1.2: *Losses inherent in the approach and their compensations.*

1.2 How it addresses the problem

The problem statement specifies eleven concrete deliverables and four hard constraints. Each is addressed by an identified component of the system, and each is verified by a specific test. The mapping below is exhaustive: no requirement is left without a mechanism, and no mechanism exists without a requirement.

What the problem statement requires	How iTantra delivers it
Lightweight, highly accurate STT and TTS for ten Indian languages	Quantised on-device acoustic and synthesis models per language, each pack under 70 MB, with word error rate measured per language and the model footprint reported
Runs locally on a low-power device	All inference on the device; no server; budgeted RAM, CPU and thermal envelope on an entry-tier handset
STT activated after detecting pauses and stoppages	Three-tier voice activity detection followed by an endpointing state machine that decides when a sentence has ended
Should form the sentences detected	Segment-level finalisation with punctuation and text normalisation, not a stream of loose words
Instantly and efficiently stream the data	A compact binary wire protocol with a ten-byte header and two-byte trailer; a full sentence in 44 bytes
Through a Wi-Fi/Bluetooth connected embedded device or another phone	A transport abstraction with four implementations, one of which is serial-shaped so it drives a LoRa radio module over Bluetooth SPP
TTS converts received text into intelligible speech, played as a voice note	Streaming chunked synthesis with playback beginning before synthesis completes, and a replayable message log
Alert messages announced at highest volume, non-interruptible	Alarm-stream audio routing, forced volume with restoration, exclusive audio focus with loss callbacks ignored, wake lock and lock-screen delivery
Two phones, one TTS one STT, verify the complete loop	A symmetric application: every device runs both halves and switches role by mode. A single application package is installed on both handsets
Work like a walkie-talkie using push-to-talk	Half-duplex mode with a large on-screen control also bound to the volume-down hardware key, plus floor state signalling
If turned off it should work like a phone	Full-duplex mode with continuous VAD-gated streaming in both directions and barge-in ducking

Table 1.3: *Requirement-by-requirement mapping to the delivered system.*

1.2.1 The four constraints

Constraint	Consequence for the design and how compliance is proved
Open source only. Proprietary, closed-source or commercial voice-activation SDKs are prohibited	No such component appears anywhere in the system. Google Speech Services, Azure Speech and Picovoice are excluded, and Google Nearby Connections is deliberately excluded from the transport layer in favour of raw platform sockets. Every dependency is licence-audited and listed in Section 5
Fully offline working. No internet-hosted API for STT or TTS	No network call at run time, ever. Language packs may be fetched once during setup; the running system never touches a network. The claim is enforced by the operating system, because the shipped manifest declares no internet permission
Approved frameworks. Open-source machine learning and TinyML frameworks; TensorFlow Lite, PyTorch Mobile or similar	The runtime is ONNX Runtime through sherpa-onnx — permissively licensed, ARM-optimised, and within the "or similar" scope. LiteRT and ExecuTorch were evaluated as alternatives and remain drop-in replacements behind the same module boundary
Low and mid-range phones. The application must run smoothly on such devices	The target device is a four-gigabyte entry-tier Snapdragon or Helio handset. Every number reported by the project is measured on that class of hardware, never on a flagship

Table 1.4: *The four constraints and the evidence of compliance.*

1.2.2 The evaluation criteria as a design driver

Eighty per cent of the assessment is allocated to three measurable properties: accuracy at forty per cent, latency at twenty per cent and efficiency at twenty per cent. This is unusual and it is a gift, because the criteria state exactly what must be instrumented. The design principle drawn from this is that the application is instrumented from the first commit: it displays a permanent latency and resource strip during operation, and it can export its own scorecard. The submission is therefore a set of measurements rather than a set of claims.

1.3 Innovation and uniqueness of the solution

Many approaches will demonstrate speech crossing a Bluetooth link. Four properties distinguish this solution, and each is difficult to assemble late but straightforward when planned from the beginning.

1.3.1 Compressing meaning rather than waveform

The central innovation is a reframing rather than an optimisation. Every existing approach to voice over constrained links attempts to compress the audio signal more efficiently. This solution discards the audio signal entirely and transmits the linguistic content, then regenerates an audio signal at the far end. Because the regenerated signal need not resemble the original waveform — only convey the same words — the information that must cross the link falls by three orders of magnitude rather than by a factor of two or three.

1.3.2 Script packing: a compression method unavailable to general systems

General-purpose text transports must use a universal encoding, because they cannot know in advance which script a message will contain. This system controls both endpoints and declares the language in the frame header, which permits a private single-byte alphabet per Indic script. This is a 3:1 gain over UTF-8 that no general-purpose messaging system can obtain, and it costs microseconds of table lookup. To the best of our knowledge, no existing radio messaging system exploits this property.

1.3.3 Cross-language operation as a free consequence

Because every device holds the template table in all ten languages, a message transmitted as a template code is spoken in whichever language the receiver has selected. A Hindi speaker's alert reaches a Tamil speaker in Tamil — with no translation model, no additional download, and no additional latency. This is not a feature that was added; it is an architectural consequence of the compression scheme. In a multilingual disaster response involving teams from different states, this property is operationally significant. The claim is stated precisely: it applies to template-coded messages, and free-form speech is delivered in the language in which it was spoken.

1.3.4 A deployment path that requires no change to the application

Bluetooth RFCOMM is a virtual serial cable, and it is the same profile that an ESP32 microcontroller or a radio modem presents. Consequently *the code that communicates with another handset is the code that communicates with a radio.* Approximately ₹1 500 of hardware and two hundred lines of firmware convert a phone-to-phone demonstration into a system with a range of two to fifteen kilometres, with no change whatsoever to the handset software. The firmware deliberately does not understand the protocol; it is a wire, and keeping it a wire is what makes it trustworthy.

```
DEMONSTRATION
  Phone A ------- Bluetooth -------> Phone B                       30 m

DEPLOYMENT
  Phone A --BT--> LoRa node ==== 865.5 MHz ====> LoRa node --BT--> Phone B
                                                               2 - 15 km
```

Figure 1.3: *The handset software is identical in both topologies. Only what sits at the far end of the Bluetooth connection differs.*

1.3.5 Evidence rather than claims

The fourth differentiator is methodological. Word error rate is reported per language at four signal-to-noise ratios rather than in clean conditions only; a separate critical-term error rate is reported with and without contextual biasing; real-time factor is reported after a thirty-minute thermal soak rather than from a cold device; and idle processor usage is taken from a system trace rather than asserted. All figures come from an evaluation harness that writes three comma-separated files, each row of which records the device and build it was produced on.

1.3.6 Why this matters specifically to the sponsoring organisation

Link type	Usable rate	Carries a voice call?	Carries an iTantra sentence?
LoRa, spreading factor 10, 125 kHz	~1 kbps	No	Yes — 0.4 s
LoRa, long-range profile	~0.3 kbps	No	Yes — 1.2 s
HF radio data link	~2 kbps	No	Yes — 0.2 s
Narrowband satellite burst	~1 kbps	No	Yes — 0.4 s
Bluetooth RFCOMM	~200 kbps	Marginally	Yes — 2 ms
Wi-Fi Direct	>10 Mbps	Yes	Yes — instant

Table 1.5: *Read the third column, then the fourth. On the four links that matter operationally — those that work when infrastructure does not — audio is impossible and iTantra is comfortable. That gap is the contribution.*

SECTION 2

TECHNICAL APPROACH

2.1 Technologies to be used

2.1.1 Platform and language

Layer	Technology	Justification
Platform	Android, minimum API 26, target API 35	The problem statement requires an Android application running on low and mid-range handsets
Language	Kotlin 2.0	Coroutines and flows map cleanly onto the queue-and-stage processing model
User interface	Jetpack Compose	Fewer files for a small team; the interface is simple and largely custom-drawn
Build	Gradle 8.10, Android Gradle Plugin 8.7	Multi-module build with enforced dependency rules
Storage	DataStore and Room	Typed configuration, and a queryable outbox for store-and-forward

Table 2.1: *Platform technologies.*

2.1.2 Machine learning runtime and models

Component	Licence	Role in the system
k2-fsa / sherpa-onnx	Apache-2.0	Principal dependency. Recognition, synthesis, voice activity detection and endpointing in one library, with an Android archive and Kotlin bindings
Microsoft ONNX Runtime	MIT	Inference engine beneath sherpa-onnx, with XNNPACK acceleration on ARM
AI4Bharat IndicConformer	Permissive	Production acoustic models covering all ten target languages
Vosk small models	Apache-2.0	Week-one prototype models; schedule insurance while export is completed
Rhasspy Piper voices	MIT	Primary synthesis voices, already in ONNX format
AI4Bharat Indic-TTS	Permissive	Higher-naturalness Indic voices where Piper quality is inadequate
Coqui TTS	MPL-2.0	Training path for languages without an adequate open voice
Silero VAD	MIT	Tier-one voice activity detection, 1.8 MB
espeak-ng	GPL-3.0 ⚠	Phonemisation. Copyleft; disclosed rather than discovered
Xiph RNNoise	BSD-3-Clause	Noise suppression on the recognition path only
Meta MMS	CC-BY-NC ⚠	Coverage gap-filler only; non-commercial, excluded from any deployability claim

Table 2.2: *Machine learning components. Two carry restrictive licences and both are disclosed here rather than left to be found.*

2.1.3 Communication and hardware

Component	Specification	Role
Bluetooth Classic (RFCOMM)	Android platform API	Default transport; a virtual serial cable
Bluetooth Low Energy (GATT)	MTU negotiated to 247 bytes	Standby transport; roughly one tenth the power
Wi-Fi	Hosted network, TCP	Reach and audio fallback
LoRa radio node	ESP32 with SX1276/SX1278, 865.5 MHz	Deployment path, licence-free band in India
Target handset	4 GB RAM, entry-tier Snapdragon 4-series or Helio G-series, Android 11+	Reference device for every reported measurement

Table 2.3: *Communication technologies and hardware.*

No third-party connectivity library is used. The transport layer is built on the Android platform Bluetooth and socket APIs alone, and Google Nearby Connections is deliberately excluded because it would breach the open-source constraint.

2.2 Methodology and process for implementation

2.2.1 End-to-end signal path

Every device runs the complete system. Nothing is a dedicated transmitter or a dedicated receiver; the role is a runtime mode, not a build variant, and one application package is installed on both handsets.

```
         SENDING DEVICE                            RECEIVING DEVICE
     +-----------------------------+           +-----------------------------+
     |  Microphone   16 kHz mono   |           |  Link receive, reassemble   |
     |             |               |           |            |                |
     |  Ring buffer, 20 ms hops    |           |  CRC, decrypt, de-duplicate|
     |             |               |           |            |                |
     |  Tier 0   energy gate       |           |  Key identifier filter      |
     |  Tier 1   Silero VAD        |  idle path |            |                |
     |             |  speech       |           |  Unpack script     -or-     |
     |  Feature extraction         |           |  expand template code       |
     |  80-dimension log-mel       |           |            |                |
     |             |               |           |  Text normalisation         |
     |  Conformer encoder          |           |  numbers, units, dates      |
     |             |               |           |            |                |
     |  Transducer decoding        |           |  Phonemisation, espeak-ng   |
     |  + contextual biasing       |           |            |                |
     |             |               |           |  VITS synthesis, chunked    |
     |  Endpoint detected          |           |            |                |
     |  400 ms / 150 ms silence    |           |  AudioTrack streaming       |
     |             |               |           |            |                |
     |  Template match    -or-     |           |  ALERT? -> alarm stream,    |
     |  script packing             |           |  maximum volume, wake lock,|
     |             |               |           |  exclusive focus, repeat    |
     |  Frame, encrypt, CRC        |           |            |                |
     +-------------|---------------+           +------------|----------------+
                   |            13 to 60 bytes              |
                   +----------------------------------------+
                     Bluetooth  ·  BLE  ·  Wi-Fi  ·  LoRa / HF
```

Figure 2.1: *Complete signal path. Both halves are present on every device.*

2.2.2 Module structure

Module	Responsibility
app	User interface, push-to-talk control, alert delivery, pairing by QR code, instrumentation display, foreground service
core-audio	Capture and playback, ring buffers, gain control, noise suppression, alert audio policy, audio focus handling
core-asr	Recogniser binding, three-tier detection, endpointing state machine, hotword and gazetteer biasing, confidence scoring
core-tts	Per-language normalisation rules, phonemiser, chunked synthesis, streaming playout scheduler
core-link	Transport interface with four implementations; discovery, reconnection, queueing
core-proto	Frame encoding and decoding, CRC-16, AES-GCM, script packing tables, template tables, sequence and replay windows, relay logic
core-models	Language pack manifest, checksum verification, resumable download, storage accounting, model lifecycle
bench	Word error rate harness, real-time factor measurement, latency logging, noise mixing, scorecard export

Table 2.4: *The eight modules and their responsibilities.*

The most important boundary in the design is that **core-proto carries no Android dependency**. The highest-risk component in the project — the frame codec — is therefore unit-tested and fuzz-tested on a desktop computer in seconds rather than on a handset in minutes, and the whole transport layer can be built and proven before any model exists.

2.2.3 Threading and lifecycle

Audio work must never touch the user-interface thread, and the recogniser must never block capture. Four threads operate with bounded queues between them.

Thread	Priority	Duty and constraint
Capture	Urgent audio	Reads 20 ms blocks, runs the energy gate and VAD, pushes speech into the recognition queue. Must not allocate memory, log, or format strings — a dropped block is a lost utterance
Inference	Audio	Drains the recognition queue, runs encoder and decoder, emits partial and final hypotheses
Link	Default	Blocking socket reads and writes, framing, retransmission timers, heartbeat
Synthesis	Audio	Chunked generation feeding the audio output in streaming mode, staying ahead of the playback clock

Table 2.5: *Threading model.*

The whole engine lives in a foreground service with a persistent notification, so that the operating system does not reclaim it while the screen is off — which is the normal operating state for a radio. Models are loaded when the service starts and held resident, so that the one to two second cold-start cost is never paid when the user presses transmit.

2.2.4 The wire protocol

Ten bytes of header and two of trailer buy priority, relay, integrity and authentication. The payload is what remains. All multi-byte fields are transmitted most significant byte first.

```
byte  0      1      2   3     4      5   6     7      8      9     10 ...   n-2 n-1
  +------+------+---------+------+---------+------+------+------+-------+---------+
  |MAGIC | TYPE |   SEQ   |FLAGS |   LEN   | SRC  |KEYID | TTL  |PAYLOAD|  CRC16  |
  | VER  | LANG |         |      |         |      |      |      |       |         |
  +------+------+---------+------+---------+------+------+------+-------+---------+
  |<----------------- header, 10 bytes ------------------->|       |<- 2 B ->|
```

Figure 2.2: *Frame format.*

Field	Bytes	Description
MAGIC / VER	1	Fixed sentinel used to resynchronise a byte stream after corruption, plus the protocol version
TYPE / LANG	1	Message type — text, alert, acknowledgement, floor control, heartbeat, template, position — and the language index from zero to nine
SEQ	2	Per-sender sequence number driving acknowledgement, duplicate suppression and the replay window
FLAGS	1	Final, partial, encrypted, fragment, packed, template, and two bits of recogniser confidence
LEN	2	Payload length; the basis of framing over a byte stream
SRC	1	Sender node identifier
KEYID	1	First byte of the hash of the shared key; a rapid rejection filter ahead of decryption
TTL	1	Remaining relay hops; decremented on forward and dropped at zero
PAYLOAD	n	Packed text, template identifier, coordinates or control data
CRC16	2	CCITT checksum over the header and payload

Table 2.6: *Frame fields.*

Addressing. Radio does not route; it broadcasts. There is no destination field: every frame reaches every device holding the shared key, exactly as a walkie-talkie operates. Selection is performed by cryptographic key rather than by address, because address filtering is a convention that any transmitter may disregard.

Integrity and authentication. The problem statement does not require this. It is included because the system can wake a locked, silenced handset and announce a message at maximum volume on command; if it accepted unauthenticated input, any transmitter within radio range could inject a fraudulent evacuation order into a disaster area. Frames are protected by AES-256 in Galois/Counter Mode with a pre-shared key held by every paired unit, and the entire header is included as associated data so that the sender identifier cannot be altered without invalidating the frame. A sliding window of sixty-four sequence numbers per sender prevents a recorded alert being replayed. The key is transferred optically by QR code and never crosses the radio medium, which is what makes a pre-shared key meaningfully pre-shared.

Reliability. Corrupt frames are discarded and never spoken, because a garbled instruction is more dangerous than a missing one. Alerts are acknowledged and retried up to three times at three hundred millisecond intervals; ordinary conversation is fire-and-forget, because retransmitting stale speech is worse than losing it. A heartbeat every two seconds distinguishes a quiet channel from a dead one, and undelivered frames queue locally and flush on reconnection.

2.2.5 Recognition pipeline in detail

Running the acoustic model continuously would exhaust the battery in roughly two hours and would fail the efficiency criterion outright. Detection is therefore tiered, so that the expensive stage runs only when the inexpensive stages agree that a human is speaking.

Tier	Mechanism	Cost per 20 ms	Purpose
0	Energy gate against an adaptive noise floor	~3 µs	Rejects true silence, which is the overwhelming majority of elapsed time. Effectively free
1	Silero VAD, 1.8 MB ONNX	~300 µs	Distinguishes human speech from door slams, engines, wind and music; prevents false wake-ups
2	Conformer acoustic model	~4 ms	Actual recognition. Runs only while tier one reports speech

Table 2.7: *Three-tier voice activity detection. With speech occupying perhaps five per cent of elapsed time, idle processor usage falls below two per cent of one core.*

Endpointing is implemented as a state machine which answers the problem statement's requirement for activation "after detecting pauses and stoppages".

```
IDLE --speech detected--> LISTENING ------------------> FINALISING --> IDLE
  ^                         |    |                          |
  |                         |    +-- 8 s elapsed -----------+   force cut, avoid
  |                         |                               |   hogging the channel
  |                         +-- trailing silence -----------+
  |                                                         |
  +--------------- frame transmitted <----------------------+
```

Figure 2.3: *Endpointing state machine.*

Parameter	Value	Rationale
Trailing silence, telephone mode	400 ms	Shorter clips natural inter-clause pauses; longer is perceptible as lag
Trailing silence, push-to-talk mode	150 ms	Releasing the key is itself an explicit end-of-utterance signal
Leading pad	250 ms	Retained from the ring buffer so the first phoneme is never clipped
Maximum utterance	8 s	Bounds latency and prevents one speaker monopolising a half-duplex channel
Minimum utterance	300 ms	Suppresses coughs, clicks and key noise

Table 2.8: *Endpointing parameters. The silence window is the largest single component of end-to-end delay, and it is a design choice rather than a computational cost — which is why push-to-talk mode is measurably faster.*

Contextual biasing. Decoding scores are boosted for a supplied phrase list. In a distress context the critical vocabulary is small and known in advance: मदद, घायल, आग, निकासी, unit call signs, sector numbers and local place names. Loading a domain lexicon and a deployment gazetteer measurably reduces error on precisely the words whose misrecognition would be most costly, and requires no retraining. Negation terms are weighted heavily, because an error that converts "do not evacuate" into "now evacuate" inverts the meaning of the message and is the most dangerous failure the system can produce.

Quantisation. Models are trained in 32-bit floating point, which is unnecessary for inference. Dynamic quantisation rewrites the weights as 8-bit integers, exploiting the fact that ARM cores execute integer arithmetic considerably faster than floating point.

Precision	Size	Relative speed	Effect on WER	Verdict
float32, as trained	~120 MB	1.0×	baseline	Unshippable
float16	~60 MB	1.3×	≈ 0	Intermediate
int8 dynamic	**~32 MB**	**2.5×**	**+1 % rel.**	**Adopted**

Table 2.9: *Four times smaller and two and a half times faster for approximately one per cent relative degradation. This single transformation is what makes on-device speech viable on an entry-tier handset.*

2.2.6 Synthesis pipeline in detail

Two of the four synthesis stages are ordinary software engineering rather than machine learning, and they account for most of the perceived output quality.

Text normalisation. No neural synthesiser handles raw digits, units or abbreviations correctly, and these are exactly what operational messages contain. A per-language rule file runs before synthesis. Distinguishing a quantity from an identifier is a rule, not a model: bare digit runs following a dialling or call-sign context are read digit-wise, while quantities followed by a unit are read as numbers.

Input	Naïve output	Correct output
112	one hundred and twelve	एक · एक · दो — digit-wise, because it is a number to dial
5 km	five k m	पाँच किलोमीटर
Sector 17	sector seventeen	सेक्टर सत्रह
14:30	fourteen colon thirty	दोपहर दो बजकर तीस मिनट

Table 2.10: *Text normalisation is the difference between a synthesiser that sounds professional and one that sounds broken.*

Phonemisation and synthesis. espeak-ng converts orthography to phoneme sequences, which are mapped to integer identifiers. Indic scripts being largely phonetic makes this stage far more reliable for the ten target languages than it would be for English. VITS then generates the waveform directly from phonemes in a single pass, including a duration predictor which produces natural rhythm, at a real-time factor of approximately 0.15 on entry-tier ARM.

Chunked streaming playout. Synthesising a complete sentence and then playing it wastes the entire synthesis duration as latency. Instead the sentence is split at clause boundaries; the first chunk is synthesised and playback begins immediately while subsequent chunks are generated against the playing clock. Total synthesis work is unchanged; perceived latency falls from approximately 700 ms to approximately 180 ms.

2.2.7 Alert delivery

The requirement that alerts be "announced at highest volume, non-interruptible" maps to a specific and verifiable sequence on Android. Each step exists because omitting it produces a silent alert in some real configuration.

Step	Mechanism	Why it is necessary
1	Route away from media using alarm usage and sonification content type	Bypasses media volume and, where configured, Do Not Disturb
2	Force the alarm stream to maximum volume before playback, and restore the previous level afterwards	A silenced handset must still announce. Restoring the prior level is mandatory and easily forgotten
3	Request exclusive transient audio focus and deliberately ignore loss callbacks	This is precisely the non-interruptible requirement
4	Hold a partial wake lock and post a full-screen-intent notification	Delivery must succeed on a locked screen
5	Reinforce with a vibration pattern, a high-contrast full-screen visual, and repetition	Redundant channels for a noisy environment or a hearing-impaired operator
6	Pre-empt the transmit queue; acknowledge and retry	An emergency must never wait behind ordinary conversation

Table 2.11: *The six-step alert delivery sequence.*

Alert-class messages additionally require explicit confirmation of the recognised text before transmission. The confirmation screen speaks the text aloud when it opens, so that it works for a non-literate operator, and the confirm and cancel targets are exactly equal in size. This is the one place where the system deliberately adds latency, and the reason is that an alert is the only message type that can cause physical harm if it is wrong.

2.2.8 Development process

The governing rule of the implementation plan is that the end-to-end loop closes before any effort is spent on model quality. A working loop with mediocre models can be improved incrementally under any amount of time pressure; excellent models with no loop cannot be demonstrated at all.

Week	Objective	Deliverable and acceptance gate
1	Capture and recognise	Skeleton application; capture into voice activity detection into a prototype recogniser; recognised Hindi text on screen, offline, on the target handset
2	Transport	Transport interface, Bluetooth implementation, frame codec with CRC, fuzz-tested decoder. Typed text traverses two handsets, independent of any model
3	**Close the loop**	Synthesis on the receiving device. **Speech in, speech out, end to end.** Baseline latency measured. The project is now de-risked
4	Production models	IndicConformer exported and quantised; five languages integrated; word error rate harness operational; the Odia gap assessed and assigned
5	Modes and alerts	Push-to-talk with the hardware key, half and full duplex, the complete alert path with lock-screen delivery
6	Latency, reach and security	Chunked synthesis, BLE and Wi-Fi transports, encryption, replay window, relay. Thermal soak testing begins
7	Coverage and hardening	Remaining languages, template tables, normalisation suites, noise pre-processing, field testing at range, multi-device relay soak
8	Evidence and rehearsal	Scorecard generation, documentation, radio-link demonstration, presentation rehearsed to a fixed script

Table 2.12: *Eight-week implementation plan with a gate per week.*

Testing is organised by layer. Everything testable without a handset is tested without a handset, because a test that requires two paired telephones will not be run often enough to catch anything. The protocol module in particular is pure JVM code, so its tests — including a fuzz harness that feeds the decoder truncated, interleaved and bit-flipped input — execute in milliseconds on a desktop. Audio and alert behaviour is tested on the target handset itself, because vendor audio policy varies enough that an emulator proves nothing.

SECTION 3

FEASIBILITY AND VIABILITY

3.1 Analysis of the feasibility of the idea

3.1.1 Technical feasibility

Every component required by the system exists today, is released under an open-source licence, and executes on ARM processors. Nothing in the design requires research; the work is one of composition and engineering. Quantised acoustic models occupy approximately thirty-five megabytes per language and synthesis voices approximately twenty-six megabytes, which fits comfortably within the storage budget of an entry-tier handset. Reference implementations exist for every part of the transport layer, in projects that are in production use today.

Capability required	Existing evidence that it is achievable
On-device Indic speech recognition	AI4Bharat IndicConformer models cover all ten languages; sherpa-onnx runs them on Android today
On-device neural synthesis	Piper VITS voices are already exported to ONNX and run in real time on mobile ARM
Low-power always-listening detection	Silero VAD at 1.8 MB is widely deployed for exactly this purpose
Text over long-range radio from Android	Meshtastic and APRSdroid are working, deployed systems
Alert delivery on a locked handset	A documented Android capability used by every alarm application

Table 3.1: *Each required capability already exists in the field.*

3.1.2 Economic feasibility

The system incurs no running cost whatsoever. There is no server to operate, no data plan to purchase, no per-message charge and no subscription. It executes on handsets that users already possess. Long-range deployment requires two radio nodes costing approximately ₹1 500 in total, each consisting of an ESP32 microcontroller and an SX1276 transceiver operating in the licence-free 865–867 MHz band allocated in India. No spectrum licence is required.

3.1.3 Operational feasibility

The operating model is that of a walkie-talkie, which is already familiar to the intended users and requires no training. The transmit control occupies at least one third of the screen and is additionally bound to the volume-down hardware key, so the system can be operated with the screen switched off and with gloved hands. Every state change is confirmed by haptic feedback and by a spoken cue rather than by a text dialogue alone, so that no part of ordinary operation requires reading.

3.1.4 Schedule feasibility

The eight-week plan is ordered so that the riskiest uncertainty is removed first. The end-to-end loop closes in week three using the weakest acceptable models, after which model quality is an upgrade path rather than a prerequisite. Prototype models are integrated in week one specifically so that the schedule survives even if the export of production models fails entirely. Transport work requires no models at all and therefore proceeds in parallel from the first day.

3.2 Potential challenges and risks

Twenty-nine failure modes have been identified and recorded, each with a designated owner, an escalation trigger and a designed response. The most significant are set out below. Severity reflects impact on the assessed criteria, not merely on the software.

ID	Risk	Severity	Consequence if unmanaged
T-01	Word error rate collapses in real acoustic noise	High	Destroys the largest single assessment criterion
T-02	Digits, call signs and place names misrecognised	High	Precisely the tokens that carry operational meaning are lost
T-03	Thermal throttling doubles real-time factor during a long demonstration	High	Latency figures measured cold do not hold in the room
T-04	Total asset size defeats the efficiency criterion	Medium	Installer exceeds the thirty-megabyte budget
T-05	No adequate open synthesis model exists for Odia	Medium	One of the ten required languages cannot be delivered
T-06	Code-mixed Hindi and English speech is mis-decoded	Medium	Realistic mixed speech degrades accuracy
T-07	Model export from the training format fails or degrades accuracy	Medium	Production models unavailable
T-08	Stream framing defect corrupts frames under load	Medium	Works on a desk, fails in the field
S-01	Fraudulent alert injected by an unauthorised transmitter	High	A false evacuation order announced at maximum volume
S-03	Recognition error inverts meaning	High	“Do not evacuate” becomes “now evacuate”
P-01	Model work consumes the schedule and the loop is unproven late	High	Nothing demonstrable at the end
P-02	Development on flagship hardware conceals performance failures	High	Reported figures do not hold on the target device

Table 3.2: *Principal risks.*

3.3 Strategies for overcoming these challenges

Each risk above has a designed response rather than an intention. The responses are listed against their risk identifiers so that the mapping is auditable.

ID	Strategy adopted
T-01	Neural noise suppression of approximately 200 KB applied to the recognition path only, together with adaptive gain, a high-pass filter and the platform noise suppressor. Critically, accuracy is *evaluated and reported at four signal-to-noise ratios* rather than in clean conditions only, so the weakness is measured rather than hidden. Template fallback engages automatically when confidence drops
T-02	Contextual biasing with a domain lexicon and a deployment gazetteer; digits displayed and confirmed on screen before transmission; a separate critical-term error metric tracked throughout the project and reported with and without biasing
T-03	Thirty-minute soak testing from week six onward; all figures reported as sustained rather than peak; thread count reduced under thermal pressure; low-power standby transport to lower the baseline temperature
T-04	int8 quantisation throughout, on-demand language packs downloaded once at setup, ABI splitting, and a base installer carrying only two languages
T-05	Identified in advance as the expected gap and assigned in week four rather than week seven. Fallbacks in order: an alternative permissive voice, then a non-commercial model disclosed and excluded from deployability claims, then a voice trained on the IIT Madras IndicTTS corpus
T-06	English retained in the biasing lexicon, and the limitation acknowledged openly with a stated roadmap rather than concealed. Honest scoping is defensible; a broken demonstration is not
T-07	Prototype models integrated in week one keep the schedule intact regardless. Export is attempted in week four with three weeks of slack behind it
T-08	A mandatory read-until-complete loop, a checksum on every frame, magic-byte resynchronisation, and fuzz testing of the decoder with truncated, interleaved and bit-flipped input. This is the single most common defect in systems of this class and is treated accordingly
S-01	AES-256-GCM with a pre-shared key distributed optically; the sender identifier lies inside the authenticated region; frames failing authentication are discarded silently
S-03	Confidence transmitted with every frame; recognised text displayed to the sender before transmission; explicit confirmation required for alert-class messages, spoken aloud so it works without reading; negation terms weighted in the biasing lexicon
P-01	**The loop closes in week three using the weakest acceptable models.** Model quality is an upgrade path, never a prerequisite. This is the single most important scheduling decision in the plan
P-02	An entry-tier target handset is acquired in week one and every benchmark is taken on it. No figure measured on a flagship is reported

Table 3.3: *Risk mitigation strategies.*

3.3.1 Graceful degradation as a general strategy

Beyond individual mitigations, the architecture is designed so that failure of any single element degrades the system rather than stopping it. If confidence falls, the system falls back from template codes to packed text, and on a sufficiently capable transport to compressed audio. If the primary transport disconnects, frames queue locally and flush on reconnection. If a peer is out of direct range, intermediate devices relay the frame. If the acoustic model cannot be exported, prototype models remain in place. If a language pack is corrupt, the system reverts to the base languages rather than crashing. Every degraded condition is displayed to the operator with a reason, because a system that silently stops working is worse than one that says it has stopped.

3.3.2 Scope discipline

A cut list is defined in advance and ordered, so that if time is lost the decision of what to drop has already been made calmly rather than under pressure. Optional benchmarking is cut first, then the audio fallback, then multi-hop relay, then cryptography, then the third transport, then languages beyond five, and last the long-range radio demonstration. The loop itself, the alert path and the measurement harness are never cut, because the first two are explicit requirements and the third is what turns the submission into evidence.

SECTION 4

IMPACT AND BENEFITS

4.1 Potential impact on the target audience

Target audience	Situation today	Impact of the solution
Disaster relief and NDRF teams	Mobile towers fail early in a flood or earthquake; handheld radios carry no voice on the links that remain	Voice-to-voice coordination continues where infrastructure has failed, in the local language, without training
Armed forces and border patrols	Narrowband tactical links; operators wearing gloves at altitude cannot type	Single-button operation with the screen off; sub-second delivery; authenticated frames
Fishermen and coastal communities	Beyond cellular range; literacy is not universal	Communication with shore in the mother tongue, requiring no reading or typing at any point
Forest, wildlife and remote medical patrols	No coverage; text messaging requires literacy and steady hands	Long-range voice-to-voice contact over LoRa with ₹1 500 of additional hardware
ISRO ground and field teams	Narrowband satellite burst channels cannot carry a voice call	Sentence delivery within a 1 kbps burst channel, which is the sponsoring organisation's own use case
Non-literate citizens generally	Every existing radio text system excludes them by design	The system is usable by a person who cannot read a single character in any script

Table 4.1: *Impact by target audience.*

The most significant impact is the one that is easiest to overlook. Every existing system that transmits messages over narrowband radio requires the sender to type and the receiver to read. That single design assumption excludes a very large fraction of the population that a disaster response is intended to serve, and it excludes them at precisely the moment when their information is most valuable. Removing that assumption is not a usability improvement; it changes who the system is for.

4.2 Benefits of the solution

4.2.1 Social benefits

- **Inclusion by construction.** The system is usable without literacy in any language. It does not merely accommodate non-literate users; it is designed so that reading never occurs in normal operation. Icons and colour carry primary meaning and text is a secondary channel.
- **Linguistic equity.** Ten Indian languages are supported as first-class citizens with their own acoustic models, voices and normalisation rules — not as translations of an English-first system.
- **Cross-language coordination.** Teams from different states can exchange template-coded alerts, each hearing them in their own language, which materially reduces coordination friction in a multilingual response.
- **Accessibility.** The same properties that serve a non-literate user also serve an operator wearing gloves, working in darkness, or with limited vision. Redundant haptic, audible and visual channels serve operators with hearing impairment.
- **Safety.** Alerts reach a locked and silenced handset at maximum volume, which is the difference between an alarm delivered and an alarm missed.

4.2.2 Economic benefits

- **Zero running cost.** No servers, no data plan, no per-message charge, no subscription and no cloud dependency.
- **Runs on existing hardware.** The application targets low and mid-range handsets that field personnel already carry, so deployment requires no device procurement.
- **Very low incremental cost for range.** Approximately ₹1 500 per pair of radio nodes converts thirty-metre operation into two to fifteen kilometres.
- **No spectrum licensing cost.** Operation is in the licence-free 865–867 MHz band.
- **No vendor lock-in.** Every component is open source and every model is replaceable behind a stable module boundary.

4.2.3 Environmental and operational benefits

- **Energy efficiency.** Three-tier detection holds idle processor usage below two per cent, supporting more than eight hours of standby listening on a single charge — significant where recharging is not possible.
- **Spectrum efficiency.** A sentence occupies 1.2 seconds of airtime instead of sixty, so many more users share the same channel, and duty-cycle regulations are respected comfortably.
- **No new infrastructure.** The system adds no towers, no base stations and no installed equipment, so it has no construction footprint.
- **Extended hardware life.** Because the system targets entry-tier devices, older handsets remain useful rather than being replaced.
- **Data minimisation.** Captured audio is never stored; recognised text is retained locally for twenty-four hours and then deleted; no data leaves the device.

4.2.4 Summary of measurable benefit

Measure	Without iTantra	With iTantra
Sentence over a 300 bps link	Impossible as voice; 60 s as Opus audio	1.2 s
Data per spoken sentence	96 000 bytes	44 bytes
Usable by a non-literate operator	No	Yes
Operable with gloves and screen off	No	Yes
Infrastructure required	Tower or cloud service	None
Recurring cost	Data plan or satellite airtime	Zero

Table 4.2: *Summary of measurable benefit.*

SECTION 5

RESEARCH AND REFERENCES

5.1 Research undertaken

The design rests on three bodies of prior work, each surveyed before the architecture was fixed: neural speech recognition for Indian languages, on-device neural speech synthesis, and text messaging over long-range low-bitrate radio. The literature contains mature solutions to each half of the problem and no solution that joins them. On-device recognition and synthesis for Indian languages are available under permissive licences and are computationally feasible on mobile hardware; long-range text messaging over narrowband radio is likewise solved with several working open-source implementations. What does not exist is a system that places speech interfaces at both ends of such a link, so that the operators never read or type. That gap is the contribution of this project.

Area studied	What was taken from it
Conformer acoustic modelling	The encoder architecture: convolution for local acoustic detail interleaved with self-attention for sentence-level context; approximately 85 % of inference cost, and therefore the target of quantisation
AI4Bharat Indic speech resources	Acoustic models and evaluation corpora specific to Indian languages, without which general multilingual models perform poorly on Indic phonology and on Odia and Kannada in particular
VITS single-pass synthesis	Waveform generation directly from phonemes including a duration predictor, which is what makes on-device synthesis fast enough and prevents metronomic output
Meshtastic firmware and Android client	Framing, addressing, relay behaviour, time-to-live handling and duplicate suppression over LoRa
Briar transport layer	Robust Bluetooth and Wi-Fi transport on Android: discovery, reconnection, degraded-state handling
APRSdroid	Android-to-radio operation over a Bluetooth serial profile — the deployment path adopted here, already in production use
Opus codec specification	Establishing the intelligibility floor of waveform coding, which is the bound the project is designed to circumvent
NIST SP 800-38D	Correct construction of authenticated encryption, in particular nonce uniqueness requirements
LoRaWAN IN865 band plan	Legal operating parameters and duty-cycle limits for the licence-free band in India

Table 5.1: *Research inputs and what each contributed.*

5.2 References

1. Indian Space Research Organisation, Department of Space, “iTantra — Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for Low Bitrate Links,” Problem Statement SIH26173, Smart India Hackathon 2026. Available: https://sih.gov.in
2. A. Gulati, J. Qin, C. Chiu, N. Parmar, Y. Zhang, J. Yu, W. Han, S. Wang, Z. Zhang, Y. Wu and R. Pang, “Conformer: Convolution-augmented Transformer for Speech Recognition,” *Proc. Interspeech*, 2020, pp. 5036–5040.
3. J. Kim, J. Kong and J. Son, “Conditional Variational Autoencoder with Adversarial Learning for End-to-End Text-to-Speech,” *Proc. 38th International Conference on Machine Learning*, 2021, pp. 5530–5540.

4. T. Javed, S. Doddapaneni, A. Raman, K. S. Bhogale, G. Ramesh, A. Kunchukuttan, P. Kumar and M. M. Khapra, “Towards Building ASR Systems for the Next Billion Users,” *Proc. AAAI Conference on Artificial Intelligence*, vol. 36, 2022.
5. K. S. Bhogale et al., “IndicSUPERB: A Speech Processing Universal Performance Benchmark for Indian Languages,” *Proc. AAAI*, 2023.
6. AI4Bharat, Indian Institute of Technology Madras, “IndicConformer, Indic-TTS, IndicSUPERB, Kathbath and IndicVoices.” Available: https://ai4bharat.iitm.ac.in
7. k2-fsa Project, “sherpa-onnx: Speech-to-text, text-to-speech and voice activity detection using ONNX Runtime,” Apache-2.0 Licence. Available: https://github.com/k2-fsa/sherpa-onnx
8. Microsoft, “ONNX Runtime: Cross-platform accelerated machine learning,” MIT Licence. Available: https://onnxruntime.ai
9. M. Hansen, “Piper: A fast, local neural text-to-speech system,” Rhasspy Project, MIT Licence. Available: https://github.com/rhasspy/piper
10. Silero Team, “Silero VAD: Pre-trained enterprise-grade voice activity detector,” MIT Licence. Available: https://github.com/snakers4/silero-vad
11. J.-M. Valin, “A Hybrid DSP/Deep Learning Approach to Real-Time Full-Band Speech Enhancement,” *Proc. IEEE International Workshop on Multimedia Signal Processing*, 2018.
12. J.-M. Valin, K. Vos and T. Terriberry, “Definition of the Opus Audio Codec,” RFC 6716, Internet Engineering Task Force, 2012.
13. M. Dworkin, “Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC,” NIST Special Publication 800-38D, National Institute of Standards and Technology, 2007.
14. LoRa Alliance, “LoRaWAN Regional Parameters, IN865–867 Band Plan,” Version 1.0.4, 2022.
15. Meshtastic Project, “Meshtastic: An open-source, off-grid, decentralised mesh network over LoRa,” GPL-3.0 Licence. Available: https://meshtastic.org
16. Briar Project, “Briar: Secure messaging over Bluetooth, Wi-Fi and Tor,” GPL-3.0 Licence. Available: https://briarproject.org
17. G. Lukas, “APRSdroid: Amateur radio APRS for Android,” GPL-3.0 Licence. Available: https://aprsdroid.org
18. A. Conneau, M. Ma, S. Khanuja, Y. Zhang, V. Axelrod, S. Dalmia, J. Riesa, C. Rivera and A. Bapna, “FLEURS: Few-shot Learning Evaluation of Universal Representations of Speech,” *Proc. IEEE Spoken Language Technology Workshop*, 2023.
19. Indian Institute of Technology Madras, “IndicTTS Database.” Available: https://www.iitm.ac.in/donlab/tts
20. Google Android Developers, “AudioAttributes, AudioFocus and Full-Screen Intent Notifications,” Android Platform Documentation. Available: https://developer.android.com